

Workgroup:	Network Working Group			
Internet-Draft:	draft-schrock-kintzele-grid-curtailment-01			
Published:	24 September 2026			
Intended Status:	Informational			
Expires:	28 March 2027			
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GRACE: Evidence-Bound Grid Curtailment Admission, Observation, and Single-Use Settlement

Abstract

This document defines GRACE, an application profile for one bounded `grid.curtailment` action. EMILIA supplies the exact-action authority and admission spine: it binds an approved objective to a pinned actuation plan, consumes that authority once in a durable admission domain, and preserves unresolved state after possible provider entry. An upstream action-specific command-authority proof is an input to that admission, and neither it nor the admission is sufficient dispatch authority alone. A deployment-pinned executor adapter returns command-channel evidence, and a separately authenticated meter supplies observed-effect evidence.

Safety trips, protective relays, emergency shutdowns, and hardwired interlocks remain outside GRACE and continue to operate when the Gate, network, identity service, or evidence service is unavailable. Missing or ambiguous post-entry evidence is preserved as indeterminate and cannot authorize blind retry.

GRACE verifies signed inputs and deterministic computations. It does not establish physical meter truth, baseline correctness, tariff eligibility, actual payment, complete mediation, or a physical grid deployment. An optional hybrid artifact-signature profile combines Ed25519 with ML-DSA-65 and requires both signatures to verify.

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1. Introduction

A curtailment instruction can cross several administrative and technical boundaries before it changes electrical load. Authentication of the dispatcher alone does not answer whether the exact event fit a customer-approved participation envelope, whether the required people approved it, whether the executor admitted it only once, or whether later measurement refers to the same action.

GRACE keeps those questions separate. It defines an exact action, a bounded envelope, a pinned actuation plan, pre-dispatch authorization and admission checks, native-command projection at the first conduit, per-device dispatch records, separately authenticated execution and measurement claims, outcome reconciliation, and single-use settlement admission. Each relying party selects and pins its own issuers, keys, policies, adapters, clocks, conduits, and state stores.

The profile does not define a demand-response tariff, a baseline methodology, a physical meter, a scheduler, a safety system, or a payment rail. Those systems remain authoritative for their own claims. GRACE binds their signed inputs to one action and records the result of deterministic verification. It does not prove that every physical effect passed through the Gate.

1.1. Requirements Language

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in BCP 14 [[RFC2119](#)] [[RFC8174](#)] when, and only when, they appear in all capitals, as shown here.

BCP 14 is indexed by the RFC Editor at [[BCP14](#)].

2. Terminology and Roles

Participation Envelope A customer-approved, finite set of power, energy, count, duration, notice, and time-window bounds. The current profile identifier is EP-FLEX-ENVELOPE-v2.

Curtailment Action The closed EP-GRACE-CURTAILMENT-ACTION-v1 object that identifies one requested power-reduction event.

Admission The executor-side decision that the exact action may enter effect custody once. Admission is not evidence that the effect occurred.

Command Authority Envelope An upstream, action-specific proof binding a principal's grant to an agent, asset, control verb, and expiry. GRACE verifies it as an input to admission; the envelope does not consume GRACE admission.

Actuation Plan A deterministic, digest-bound projection of the approved curtailment objective into ordered device steps. The plan is not a dispatch acknowledgment or effect observation.

Native Command Projection The canonical physical-act fields and conduit-established context derived for one plan step at the first conduit. Exact request bytes are retained separately as dispatch evidence.

First Conduit The earliest deployment-pinned boundary at which a GRACE plan step becomes a device- or protocol-specific command.

Durable Admission Domain The authoritative state domain that atomically reserves, enters, and terminalizes one action across every redundant route, worker, handoff, and restart in the protected deployment.

Actuator The deployment-pinned adapter that attempts the physical or scheduler operation and signs an acknowledgment.

Provider Outcome The command-channel result reported by the executor or downstream provider. It does not establish the physical effect.

Meter A separately pinned observation source. A meter statement is a signed source claim, not proof of physical truth or calibration.

Independent Readback An authenticated observation from a relying-party-pinned source in a control domain distinct from the command-channel source. A declared control-domain identifier is policy input, not proof of organizational independence.

Observed-Effect Relation The deterministic relation between an accepted observation and the effect bound in the approved action, such as `in_bounds`, `out_of_bounds`, `divergent`, or `unresolved`.

Safety Function A trip, protective relay, emergency shutdown, or interlock whose ability to reach a safe state does not depend on GRACE admission.

Settlement Effect A deployment-supplied downstream operation invoked only after the required evidence reconciles. GRACE controls at-most-one invocation attempt for one entitlement key. It does not define or prove payment.

3. Trust and Deployment Inputs

A conforming deployment **MUST** independently configure the accepted action, envelope, plan, and native-projection versions; approver roster and threshold; WebAuthn relying-party identifier and origins; accepted upstream command-authority profiles; actuator and meter keys; source roles and control-domain identifiers; validity and freshness policy; permitted devices, conduits, and command families; durable admission-domain identifier; Action State signing key; execution store; settlement store; and settlement adapter. Presenter-supplied trust anchors or policy values **MUST NOT** influence acceptance.

When a deployment uses a proof from [\[OT-COMMAND-AUTHORITY\]](#), it **MUST** verify that proof under relying-party policy and evaluate its freshness before admission. The proof **MUST** bind the signed grant to the specific agent, asset, control verb, and expiry and **MUST** be checked for coverage against this specific action, not merely against the breadth of the grant. It establishes who asked and what that principal authorized the agent to ask for. GRACE then admits that exact request once and consumes the admission. Both bindings are action-specific; neither is sufficient alone. The upstream proof does not consume GRACE admission and does not establish execution or physical effect.

The deployment **MUST** maintain an effect-path inventory covering every route capable of reaching the controlled devices. Every route for which the deployment claims GRACE enforcement **MUST** use one configured durable admission-domain identifier and one authoritative admission state domain. Redundant ingress, replacement workers, handoffs, and restarts **MUST NOT** mint fresh authority.

The effect-path inventory **MUST** enumerate paths by the conduit that reaches the device, not only by the application intending to use it. Engineering laptops, vendor maintenance tooling, local HMI panels, and serial drops behind a gateway are separate paths when they can reach the device. An inventory containing only routes already mediated by GRACE is incomplete.

Every protected effect path for which a deployment claims GRACE enforcement **MUST** pass through the same admission boundary. An implementation that can be bypassed may still emit evidence, but it **MUST NOT** claim complete enforcement. A receipt, log entry, or transparency inclusion cannot establish the absence of an unmodeled bypass path.

4. Curtailment Action

The action **MUST** be a closed JSON object [RFC8259] that can be serialized with JCS [RFC8785]. Unknown, duplicated, missing, or non-canonicalizable members are a refusal. Its required members are:

```
{
  "@version": "EP-GRACE-CURTAILMENT-ACTION-v1",
  "action_id": "grace:event:...",
  "action_type": "grid.curtailment",
  "effect_class": "power_reduction",
  "facility": "facility:...",
  "target_delta_kw": "18000",
  "window": {
    "not_before": "2026-07-15T20:15:00.000Z",
    "not_after": "2026-07-15T21:45:00.000Z"
  },
  "issued_at": "2026-07-15T20:00:00.000Z",
  "expires_at": "2026-07-15T21:45:00.000Z",
  "baseline_method_hash": "sha256:...",
  "control_mode": "human_on_the_loop",
  "envelope_id": "grace:envelope:...",
  "requested_by": "ep:agent:grid-coordinator"
}
```

target_delta_kw **MUST** be a positive decimal string with at most three fractional digits. Timestamps **MUST** be UTC RFC 3339 timestamps with millisecond precision. issued_at **MUST** precede window.not_before. expires_at **MUST** equal window.not_after. control_mode is either human_on_the_loop or human_in_the_loop. The baseline-method digest commits to the program-selected method; it does not validate that method.

The verifier **MUST** derive the action digest and CAID [EP-CAID] from the exact controlled-action projection. A supplied identifier **MUST NOT** replace that derivation.

5. Participation Envelope and Containment

The action names one EP-FLEX-ENVELOPE-v2 envelope. The relying party **MUST** obtain that envelope from its pinned authority and **MUST** verify that it applies to the facility, program, and evaluation time. The bounds object contains positive max_event_mw, max_period_mwh, max_events, max_event_hours, min_notice_minutes, and a participation window.

Before dispatch, the executor **MUST** compare power with power, energy with energy, count with count, and duration with duration. The action's MW cannot exceed max_event_mw. Its MW multiplied by window duration cannot exceed the remaining period MWh budget. One more event and its duration must fit the remaining event and event-hour budgets. Its notice and window must fit the envelope.

A required bound or present spent-accounting value that is missing, negative, or unparseable is a refusal. An omitted spent value asserts zero prior settled usage; deployments that cannot make that assertion **MUST** return indeterminate rather than use zero.

For native-device dispatch, the relying-party policy **MUST** also identify the participating device set, allowed native command families, permitted conduits, and plan validity interval. A plan step for a device, command family, conduit, or time outside that participation boundary **MUST** be refused before provider entry.

6. Human Authorization

GRACE does not define credential-to-person or credential-to-device enrollment. The relying party **MUST** state the verified identity-layer result and freshness policy it requires. Without that result, GRACE may report that a configured credential signed an input, but it **MUST NOT** infer a natural person from the credential alone.

The relying party selects an ordered approver roster and a threshold using the action-bound authorization evidence in [EP-RECEIPTS]. Each accepted approval **MUST** be a Class-A WebAuthn [WEBAUTHN] signoff over a context that binds the controlled action digest, actuation-plan digest, presentation digest, policy digest, initiator, approver, roster index, threshold, validity window, relying-party profile, app, platform, credential, and device key.

The verifier **MUST** require user verification, the pinned relying-party identifier and origin, distinct approvers, distinct device and credential identifiers, initiator exclusion, the exact roster index, and the selected threshold. The quorum semantics are those of [EP-QUORUM]. A valid signature that does not match the relying party's current roster, policy, action, presentation, or ceremony window is a refusal.

7. Actuation Plan and Native Command Binding

A deployment that projects one curtailment action into one or more device commands **MUST** derive a deterministic actuation plan before admission. The plan **MUST** bind the exact action digest, envelope digest, plan validity interval, participating devices, ordered steps, and the configured durable admission-domain identifier. The complete plan **MUST** be canonicalized and hashed. Human authorization and the admission record **MUST** bind that plan digest.

Each plan step **MUST** contain at least a step identifier, device identifier, conduit identifier, command-family identifier, native command digest, security-context digest, and order within the plan. The native command digest covers the complete, canonically encoded set of material fields and conduit-established context defined by the pinned transport binding. The security-context digest covers the security-relevant addressing, protocol, authentication, role, and transport context used to interpret the request. A per-attempt native request digest separately covers the exact bytes sent.

```

{
  "@version": "EP-GRACE-ACTUATION-PLAN-v1",
  "plan_id": "grace:plan:...",
  "action_id": "grace:event:...",
  "action_digest": "sha256:...",
  "envelope_digest": "sha256:...",
  "admission_domain_id": "grace:admission-domain:...",
  "validity": {
    "not_before": "2026-07-15T20:15:00.000Z",
    "not_after": "2026-07-15T21:45:00.000Z"
  },
  "steps": [{
    "step_id": "step:1",
    "order": 1,
    "device_id": "device:...",
    "conduit_id": "conduit:...",
    "command_family": "deployment-pinned-family",
    "native_command_digest": "sha256:...",
    "security_context_digest": "sha256:..."
  }]
}

```

The plan and native command are separate bindings. The plan is the protocol-independent projection of the approved objective. The native command is derived at the first conduit under one pinned transport binding. Binding only the plan leaves the device operation unbound; binding only the command leaves the approved objective unbound.

A transport binding **MUST** classify every field in a native request into exactly one of four classes: material, conduit context, correlation only, or fixed or derived. Material and conduit context contribute to the native-command digest. Correlation-only and fixed-or-derived fields do not. A field that cannot be classified, or a malformed request, **MUST** be refused before admission and before evidence lookup.

Device identity **MUST** come from conduit-established link or session context, not from a sender assertion on the wire. A wire value inconsistent with that context **MUST** be refused without resolving the disagreement in favor of either source. A Modbus unit identifier of zero **MUST** be refused by a device-scoped binding because it may denote broadcast rather than one device.

The classification binds the physical operation, not a display label or the incidental number of octets carrying it. A Modbus 4xxxxx or 0xxxxx label **MUST NOT** replace the zero-based protocol address. Modbus function 0x06 and function 0x10 with quantity one encode the same single-register write through different field sets and therefore yield different native-command digests. A conduit **MUST NOT** normalize either form into the other, and authority bound to one form **MUST NOT** admit the other. DNP3 requests that vary only in an unbound qualifier encoding **MUST** identify the same operation and yield the same native-command digest when the pinned binding decodes them to the same complete field set. The complete DNP3 control octet, operation count, on-time, and off-time are material and **MUST** be bound.

Two-phase DNP3 SELECT and OPERATE and multi-object requests are outside this revision. A later profile admitting them **MUST** bind both phases to one authorization and state machine, or the complete ordered object set to one digest, respectively.

Encoding, projection, transport acceptance, and authority to dispatch are separate claims. A serializer or conduit that successfully produces or accepts native bytes does not authorize those bytes. Mutation of a material action field, plan field, classified native-operation field, or security-relevant context after authorization

MUST cause a refusal. A correlation-only field may vary without changing the authorized physical act, but it **MUST NOT** serve as the attempt reference.

This revision deliberately leaves protocol-specific Modbus, DNP3, and vendor command-field profiles to separate deployment bindings. Such a binding **MUST** identify every material field and its canonical byte representation; it **MUST NOT** weaken the generic binding above.

8. Executor Admission and Refusal Semantics

The executor applies the Action Evidence Boundary [EP-AEB] in this order:

1. Validate the closed action and deterministic actuation plan and require both to be active.
2. Derive and verify action-to-envelope and plan-to-participation containment.
3. Derive the expected outcome policy and require its digest to be inside the human-approved policy bytes.
4. Verify the human authorization over the exact action and plan digests against the pinned profile.
5. Verify every required upstream action-specific command-authority proof, including grant coverage, expiry, and freshness, under relying-party policy.
6. Require the pinned first conduit, executor adapter, meter, Action State signer, admission store, settlement store, and settlement adapter.
7. Atomically reserve the exact execution idempotency key before provider entry.

The execution idempotency key **MUST** bind the action identifier, action digest, actuation-plan digest, and configured durable admission-domain identifier. Every ingress path, executor instance, handoff, and restart **MUST** use the same authoritative store. The authority-spend record and provider-attempt record **MUST** be created together or not at all in one durable, atomic transaction. Executor ownership **MUST** be fenced so that a losing owner cannot complete provider entry after the fence moves; a lease timeout alone is insufficient. Concurrent reservation failure is `refuse_replay`. Storage unavailability is a refusal before provider entry.

A deployment **MUST** distinguish a failure known to have occurred before provider entry from a failure after provider entry may have occurred. Once provider entry may have occurred, a lost, malformed, conflicting, or exceptional result **MUST** produce `execution_indeterminate`. The executor **MUST** preserve or burn the reservation, set `retry_safe` to false, and **MUST NOT** invoke the same admission again. Authenticated reconciliation is required. A later attempt is a new action with new authority.

The `execution_indeterminate` state is GRACE's mapping of the unresolved state in [OT-COMMAND-AUTHORITY]. The consumed authority and provider-attempt record **MUST** survive a Gate or enforcement-point restart; a power cycle **MUST NOT** restore consumed authority to unused.

Pre-invocation failures use a mechanism-named refusal such as `refuse_action`, `refuse_action_not_active`, `refuse_outside_envelope`, `refuse_outcome_policy`, `refuse_authorization`, or `refuse_adapter_unavailable`. Post-invocation evidence failures use `effect_unconfirmed` and are not retry authorization.

9. Per-Device Dispatch Records

The dispatch request **MUST** bind the complete action, action digest, envelope digest, actuation-plan digest, plan-step identifier, native-command digest, security-context digest, human-authorization digest, execution idempotency key, and operator.

For every device attempt, the implementation **MUST** preserve the operation, admission, plan-step, device, conduit, and attempt identifiers; native-command, exact native-request, and security-context digests; admission and provider-entry times; command-channel acknowledgment state; provider outcome and evidence digest; and, when available, the observed-effect relation, evidence digest, and observation time. One aggregate success value **MUST NOT** hide a missing, failed, or unresolved device attempt.

The attempt identifier **MUST NOT** be a protocol correlation field. In particular, an SNMP request-id, Modbus MBAP transaction identifier, or DNP3 application-control sequence number is client-chosen and reusable and does not identify any part of the physical act.

A COSA acknowledgment, or an acknowledgment returned by another deployment-pinned executor adapter, **MUST** bind the event, action, plan, step, dispatch-request, idempotency key, executor identity, provider outcome, and dispatch time, and **MUST** verify under the pinned executor key. It is downstream evidence about a dispatch that EMILIA has already admitted. It does not authorize the action and does not establish that electrical load changed.

A native function that defines no application response is an ordinary protocol case, not proof of failure. After a DNP3 DIRECT_OPERATE_NR request has crossed provider entry, the attempt is indeterminate unless independent authenticated reconciliation establishes the effect. The authority remains consumed and the request **MUST NOT** be retried under the same admission.

10. Meter Statement and Compliance Computation

The meter statement **MUST** bind the meter identifier, event identifier, action digest, exact action window, unit, baseline MW, ordered sequence of timestamped load intervals, measurement class, observation time, and signer key identifier. It **MUST** verify under a separately pinned meter key.

For an observation to be used as independent readback, its source **MUST** be in a relying-party-pinned control domain distinct from the command-channel source. A status object returned by the same management card, executor, or command conduit is command-channel evidence, even when it uses a different object identifier. It **MUST NOT** be relabeled as independent effect evidence.

The meter statement **MUST NOT** contain `baseline_method_hash`. Program and tariff rules remain in the authorized action and relying-party policy. This separates a signed measurement-source claim from the rule used to interpret it.

The reference computation subtracts average interval load from the signed baseline MW, divides the delivered MW by ordered MW, and marks the event compliant at a profile-selected threshold. A deployment **MUST** pin the computation and threshold it uses. The result states what follows from the accepted inputs. It does not prove the baseline was economically correct, the readings were physically true, or the event qualifies under a tariff.

11. Partial Execution and Outcome Binding

The action's human-approved policy binds source-routed predictions for an executor observation and an independent-observer meter observation. Each Outcome Observation [EP-OUTCOME] **MUST** bind the authorization record, action digest, CAID, consumption nonce, operation identifier, facility, observation window, source role, source class, and observed effect.

The relying party **MUST** pin source keys, roles, classes, control domains, status and validity, distinctness requirements, observation windows, and maximum attestation delay. A declared control domain is relying-party policy input, not proof of organizational independence.

An independent observation is admissible only when its control domain is distinct from the command channel, it verifies under its own pinned key, and its observation time is inside the admitted window. A second object read from the same communications card is in the same control domain and **MUST NOT** satisfy independent readback. An inadmissible observation and an admissible observation that disagrees are different results and **MUST NOT** share one stored state.

Each device attempt **MUST** retain one of five states: `not_entered`, `invoking`, `committed`, `proven_not_committed`, or `indeterminate`. A device moves to `invoking` at provider entry. `committed` requires both command-channel evidence and an independently authenticated admissible observation. `proven_not_committed` requires positive evidence from the sending host that the operation did not leave it; absence of an acknowledgment is insufficient. Every other post-entry result is terminal `indeterminate` for that attempt.

The command-channel statement and independent observation **MUST** be stored separately. Reconciliation writes a third record and **MUST NOT** overwrite either source record. It may resolve the attempt; it never grants permission to redispatch.

A restart or ownership handoff while any device is `invoking` or `indeterminate` **MUST NOT** create new dispatch authority. The replacement owner inherits the existing attempt, first reads the durable admission and attempt history, and does not open a second attempt. A new admission for the same action while an earlier attempt remains `indeterminate` is refused by default. A deployment that deliberately widens that rule **MUST** record the widened policy and the new authority; it cannot reuse the first admission.

The aggregate result **MUST** be derived from the complete per-device matrix. While any device remains `indeterminate`, the aggregate is `indeterminate`. A command-channel acceptance with an out-of-bounds admissible readback is `effect_observed` and `out_of_bounds`, not `effect_confirmed`. A command-channel rejection with an in-bounds admissible readback is `divergent`. Missing, stale, wrong-window, same-domain, or otherwise inadmissible readback is `effect_unconfirmed`. These states **MUST NOT** be collapsed into a generic success, failure, or retry authorization.

The minimum control-room view **MUST** show, per device: what was requested in plant terms; the command-channel result and time; the independent observation and whether it was admissible; each failed admissibility condition; the five-state device result; any observed interlock action; and any requested operator action. Unknown values **MUST** be shown as unknown rather than as zero or blank.

Any summary label shown to an operator **MUST** be derived from the underlying facts at display time and **MUST NOT** replace those facts in storage.

Settlement admission requires a reconciled, committed result with the profile-selected in-bounds relation. No other state may reach the settlement adapter.

12. Independent Safety Boundary

Trips, protective relays, emergency shutdowns, branch protection, and hardwired interlocks are outside the GRACE authority and admission path. The upstream command-authority proof [\[OT-COMMAND-AUTHORITY\]](#) does not alter that safety boundary. A GRACE implementation **MUST NOT** delay, suppress, consume, or

condition authority required by an independent safety function. The safety function **MUST** remain capable of reaching its safe state while the Gate, network, identity service, evidence service, or executor adapter is unavailable.

A safety action may be observed and recorded after the fact. Such a record **MUST NOT** claim that GRACE admitted the safety action or count it as admitted curtailment. A safety trip during an unresolved dispatch also **MUST NOT** authorize replay of the original admission.

GRACE cannot establish that a protective path is independent. A deployment claiming conformance **MUST** declare each protective path and demonstrate its operation with the Gate, admission store, identity services, and network unavailable. Routing a protective function through the Gate merely to record it violates this section.

13. Action State Signed Statement

Before provider entry, a deployment policy may require a signed audit statement over the exact action, plan, authorization, admission, and first-conduit projections. Registering that statement with a transparency service does not authorize dispatch and does not establish later execution or effect.

After outcome reconciliation, the profile emits a COSE_Sign1 [RFC9052] Action State Signed Statement using [ACTION-STATE] format version 2 and media type `application/agent-action-capsule+json`. The statement binds the action, actuation plan, admission domain, operator, developer, authorization digest, per-device dispatch records, meter digest, constraints evaluated, disposition, and confirmed or unresolved effect.

The capsule identifier is derived from canonical capsule bytes. The JSON wrapper, COSE payload, protected headers, key identifier, and statement digest **MUST** cross-check. The current reference output is an `unregistered_signed_statement`. It is not a SCITT transparency-service registration or proof of ledger inclusion. When a deployment adds SCITT inclusion evidence, that evidence proves only the statement's inclusion under the transparency-service policy.

14. Single-Use Settlement Admission

The one-time entitlement key is the injective serialization of `[envelope_id, event_id, meter_payload_digest]` under profile `EP-GRACE-SETTLE-v1`. Every member is required. The meter digest **MUST** be a lowercase SHA-256 digest of the accepted signed meter body.

Only a compliant computation and an in-bounds reconciled outcome may reach the settlement adapter. The settlement store **MUST** reserve the entitlement atomically before invocation. A duplicate is `settlement_already_consumed`. If invocation begins and its result is unknown, the implementation **MUST** commit or preserve the reservation and **MUST NOT** issue another settlement attempt under the same entitlement.

This is at-most-one admission to the configured settlement effect in one authoritative state domain. It is not exactly-once physical payment, global double-spend prevention across independent authorities, tariff eligibility, or evidence that funds moved.

15. Proof-of-Curtailment Bundle

The EP-GRACE-PROOF-OF-CURTAILMENT-v1 bundle is a signed, offline-verifiable summary. It binds the action and action digest, envelope digest, baseline-method digest, actuation plan and digest, authorization digest, admission record, native-command projections, per-device attempts, provider-evidence digests, readback-evidence digests, complete device-state matrix, aggregate Outcome Binding result, deterministic compliance result, Action State statement digest, and settlement admission result. Unavailable evidence **MUST** remain explicit; it **MUST NOT** be omitted in a way that implies success.

Bundle verification establishes integrity and binding relative to the verifier's pinned trust and policy inputs. It does not independently establish any source claim as real-world truth.

16. Artifact Signature Profiles

The baseline profile signs canonical artifact bytes with Ed25519. The optional EP-GRACE-ARTIFACT-SIGNATURE-v2 profile uses a fixed required algorithm list, in order: Ed25519 and ML-DSA-65 [FIPS204]. Its signing bytes commit to the complete artifact and the required algorithm list.

A v2 verifier **MUST** require exactly one signature under each independently pinned key and **MUST** accept only when both signatures verify. Missing algorithms, reordered or narrowed requirements, key substitution, malformed signatures, or an unavailable ML-DSA verifier are refusals. A v1 verifier **MUST NOT** silently accept a v2 artifact.

Hybrid signing provides cryptographic migration evidence. It does not make a software-held key hardware-backed, FIPS validated, deployed, or operationally independent. A deployment claiming the hybrid profile **MUST** identify and protect both actual signing keys.

17. Privacy Considerations

Curtailment records can reveal facility identity, capacity, operating schedules, energy consumption, human approvers, device identifiers, and commercial program terms. Implementations **SHOULD** minimize portable fields, disclose raw intervals only to parties that require them, use digests when independent recomputation is not required, and apply retention and access policies appropriate to critical-infrastructure data.

Selective disclosure or encryption cannot repair an overbroad signed source object. Deployments **SHOULD** separate public verification material from restricted operational evidence and preserve enough binding information to detect substitution.

18. Security Considerations

The principal attacks are action substitution, envelope overflow, actuation-plan or native-command substitution, stale approval, initiator self-approval, duplicate approvers, split admission domains, bypass paths, redundant-route or replacement-worker replay, executor ownership races, crash or restart after possible provider entry, partial actuation, lost acknowledgment, actuator-key substitution, stale or conflicting readback, meter-key substitution, market-rule smuggling into meter data, observation-source collapse, accidental capture of an independent safety path, wrapper substitution, signature-leg stripping, and duplicate settlement admission. The preceding sections require fail-closed checks or explicit indeterminate states for each.

Key compromise remains a trust-root failure. Historical signatures can establish what a key signed; they do not prove who controlled the key or that the signed claim was true. A separately keyed meter under the same operator's control may improve fault isolation but does not create organizational independence.

GRACE cannot prove that the selected baseline is fair, a meter is calibrated or untampered, a scheduler has no bypass, a declared control domain is honest, a payment rail moved funds, or an authority's upstream population is complete. Deployment claims **MUST** state those acceptance roots.

19. Non-Normative RDU101 Worked Example

This section is an implementation sketch supplied for review. It is not a protocol requirement and does not report an energized actuation test. The cited hardware characterization is read-only: TEST-01 is complete, TEST-02 remains strictly on hold, and no outlet was de-energized during this work.

The example device is a Vertiv Liebert GXT5-3000LVRT2UXLN, rated 2700 W, 3000 VA, and 120 V AC, with NEMA 5-20R programmable receptacles. A Liebert RDU101 management card running firmware 1.9.3.0 exposes LIEBERT_GP_PWR.MIB over SNMPv3 AuthPriv using SHA-1 authentication and AES-128 privacy. The proposed action is to de-energize programmable receptacle group 2 through point 4365, lgpPwrOutputReceptacleControlEntry, at OID 1.3.6.1.4.1.476.1.42.3.9.20.1.20.1.2.1.4365.2. Four rows, indices 1 through 4, were present during characterization. The outlet-group identifier is a different object and is not the command target.

The worked-example topology consumes the action-specific proof defined by [OT-COMMAND-AUTHORITY] before EMILIA admission and keeps independent observation and safety outside the command path.

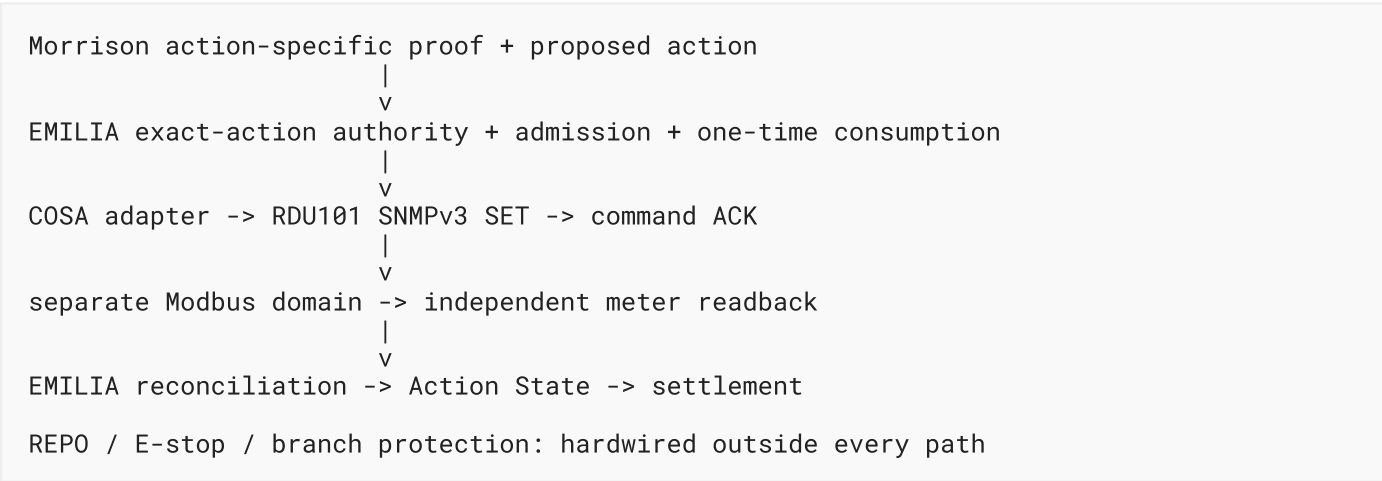


Figure 1: Example control and evidence order

A Command Authority Envelope conforming to [OT-COMMAND-AUTHORITY] establishes who asked and what that principal authorized the agent to ask for, bound to this action, asset, control verb, and expiry. EMILIA evaluates that proof as an upstream input, binds the exact action and RDU101 projection, and admits and consumes that exact request once before provider entry. Both bindings are per-action and neither is sufficient alone. The COSA adapter then issues the native SNMPv3 operation and returns the RDU101 response as downstream command-channel evidence. An SNMP SetResponse means that the management card processed the SET. It does not establish the electrical state of the receptacle or the attached load.

The independent readback in this example comes from an instrumented Modbus power meter through a separate channel and control domain from the RDU101. A second object read through the RDU101 remains same-domain command-channel evidence and is not independent readback. The hardwired REPO input, emergency stop, and branch overcurrent device remain outside the upstream command-authority proof [OT-COMMAND-AUTHORITY], EMILIA, and COSA and continue to operate without them.

Nominal An authenticated accepted acknowledgment plus independent readback within the approved bound yields `in_bounds`, `effect_confirmed`, and `invoked_confirmed` in the EP-GRACE-PROOF-OF-CURTAILMENT-v1 bundle.

Degraded communications A transport timeout or unavailable channel after possible provider entry yields `degraded_transport_timeout`, `reason_channel_unavailable`, `effect_unconfirmed`, `retry_safe=false`, and `not_invoked_refused`.

Missed bound or discrepant readback An accepted acknowledgment with admissible out-of-bounds readback yields `state_mismatch`, `out_of_bounds`, `effect_observed`, and `not_invoked_refused`, not `effect_confirmed`.

Conflicting telemetry A command-channel result and admissible readback that conflict yield `state_mismatch`, `divergent`, `effect_divergent`, and `not_invoked_refused`.

Unresolved readback Missing, stale, same-domain, or otherwise inadmissible readback yields `state_mismatch`, a null observed-effect relation, `effect_unconfirmed`, and `not_invoked_refused`.

Safety interlock trip A hardwired interlock trip is recorded as `safety_interlock_tripped`, `effect_unconfirmed`, and `not_invoked_refused`. It is not an in-bounds GRACE effect, and the same admission is not retried.

Every non-nominal path stops the same admission. There is no ambient or blind retry across degraded communications, disagreement, unresolved readback, or an independent interlock trip. A subsequent attempt requires a new action, new authorization, and new admission.

20. Conformance Cases

A conformance suite for the -01 requirements **MUST** cover at least:

1. exact objective, plan, classified native-command, and native-request binding;
2. refusal when a material native-operation field or security-relevant context changes;
3. concurrent duplicate arrival through two ingress paths sharing one durable admission domain;
4. a replacement worker attempting the same operation without being granted fresh authority;
5. a restart known to occur before provider entry;
6. a lost response after provider entry, preserving consumed authority and `retry_safe=false`;
7. partial execution across multiple devices;
8. admissible command acknowledgment and readback that conflict;
9. authenticated reconciliation that resolves an attempt without redispatch;
10. operation of the declared protective path with the Gate, admission store, identity services, and network unavailable;
11. a new executor instance inheriting an indeterminate attempt without second entry or fresh authority;
12. same-domain readback recorded as inadmissible and distinguished from admissible readback that disagrees; and

13. an aggregate that remains indeterminate while any one of several devices is unresolved.

The staged packet validates document integrity and claim boundaries. It does not claim that these thirteen behavioral cases have been implemented or passed.

21. Implementation Status

This section records implementation evidence and may be removed before publication as an RFC.

The public EMILIA repository [[GRACE-IMPLEMENTATION](#)] contains a TypeScript reference circuit for exact action construction, two distinct Class-A approvals, bounded admission, a COSA-labeled simulated actuator, a separately keyed simulated meter, Outcome Binding, an Action State Signed Statement, and one-time settlement admission. The adapters are marked `simulation: true`. The circuit is not evidence of a physical grid event, production COSA integration, utility adoption, meter truth, or an actual payment.

The published -00 packet recorded 80 targeted tests across four files and six declared vertical vectors [[GRACE-VECTORS](#)]. Those are same-team reference results, not an independent implementation or interoperability claim. They do not validate the new -01 actuation-plan, first-conduit, per-device, durable-domain, safety-boundary, or RDU101 requirements.

This -01 text is a staged review candidate. Its companion validation checks document integrity and the required claim boundaries. No energized actuation test, production COSA integration, utility deployment, or physical meter validation has been performed for this revision.

22. IANA Considerations

This document has no IANA actions. The `grid.curtailment` and `EP-GRACE-*` identifiers are profile-local strings and are not entries in an IANA registry.

23. Normative References

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- [**EP-CAID**] Schrock, I., "Canonical Action Identifier: Stable Identity for Material Agent Actions", Work in Progress, Internet-Draft, draft-schrock-canonical-action-identifier-02, August 2026, <<https://datatracker.ietf.org/doc/draft-schrock-canonical-action-identifier/>>.
- [**EP-OUTCOME**] Schrock, I., "Outcome Binding for High-Risk Agent Actions", Work in Progress, Internet-Draft, draft-schrock-ep-outcome-binding-00, July 2026, <<https://datatracker.ietf.org/doc/draft-schrock-ep-outcome-binding/>>.

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- [EP-RECEIPTS]** Schrock, I., "Authorization Receipts for High-Risk Agent Actions", Work in Progress, Internet-Draft, draft-schrock-ep-authorization-receipts-12, August 2026, <<https://datatracker.ietf.org/doc/draft-schrock-ep-authorization-receipts/>>.
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24. Informative References

- [GRACE-IMPLEMENTATION]** EMILIA Protocol, "GRACE Reference Implementation", 2026, <<https://github.com/emiliaprotocol/emilia-protocol/tree/main/lib/grace>>.
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